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Physics-informed multi-gated convolutional recurrent network for extreme wind speed prediction

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HIGHLIGHTS

- Physics-Based Volatility Proxy from pressure-gradient force discrepancy.
- Membership-Gated Convolutional Recurrent Module (MG-CRM) routes updates by regime.
- Differentiable Evolutionary Mixture (DEM) layer performs in-network feature selection.
- Volatility-Adaptive Hybrid Loss (VAH-Loss) prioritizes extreme-event accuracy.

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ABSTRACT

Accurate wind speed prediction (WSP) is critical for maintaining grid stability and economical operation of wind-integrated power systems due to the cubic relationship between wind speed and power generation. However, current deep learning models often employ static architectures and loss functions that do not adapt to atmospheric volatility, which leads to large power and energy errors during extreme events. We present the Volatility-Regulated Temporal Evolutionary Network (VORTEX), a physics-informed deep learning framework for extreme WSP in operational settings. The method incorporates a physics-based volatility proxy derived from the pressure-gradient force discrepancy to regulate learning. The model uses a Membership-Gated Convolutional Recurrent Module that trains different sets of weights for different volatility conditions. Furthermore, a differentiable evolutionary mixture layer refines the learned features before the final prediction and optimizes feature representation. We evaluate the method on four geographically diverse locations, namely Dodge City (United States), Sarmiento (Argentina), Jiuquan (China) and Pincher Creek (Canada). Across the 6-hour forecasting horizon, VORTEX consistently achieves the best overall performance on all four datasets, reducing RMSE by 4.33%–9.23% and MAE by 4.59%–10.28% relative to the strongest baseline. Its advantage further widens under the most volatile conditions, with Q4 RMSE reductions of up to 13.30%, and the improvements remain statistically significant across all datasets. At the Jiuquan site, it reduces RMSE by 7.64% and MAE by 6.75% relative to the best baseline, indicating tangible benefits for short-term dispatch and reserve scheduling.

1. Introduction

Electric power systems worldwide are increasing their share of renewable energy sources, aligning with the targets of Sustainable Development Goal 7 (SDG-7) on affordable and clean energy [1]. Yet, recent studies indicate that current deployment rates remain insufficient to meet these targets [2]. A primary challenge is the inherent intermittency of renewable sources, which introduces grid instability risks, raises balancing costs and can force emergency load curtailment. Wind power faces a uniquely severe intermittency problem governed by a nonlinear scaling law. The cubic relationship between wind speed and turbine output means that a small input variation produces a disproportionately large output variation [3]. This becomes even more critical during periods of extreme volatility, where wind speed changes rapidly over short

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time intervals and the observed acceleration departs markedly from that expected from the large-scale pressure-gradient force. Such periods are especially difficult to predict because gust ramps shift the pattern faster than learned dynamics adapt and these rare cases occupy only a small fraction of the dataset, which biases most data-driven models toward calm regimes and produces large power errors that undermine near-term dispatch and reserve scheduling.

1.1. Existing research

Existing wind speed prediction (WSP) methods are generally classified into four categories, namely physical, statistical, machine learning and deep learning methods. A more recent trend is the development of hybrid models, which combine two or more of these approaches to leverage complementary strengths and improve predictive accuracy in operational settings. Physical models for WSP employ Numerical Weather Prediction (NWP) systems, solve the governing equations of atmospheric physics to simulate future atmospheric states and thus provide a principled foundation for WSP [4]. Physical methods advance through high-resolution limited-area models such as the Weather Research and Forecasting system, which can be configured at kilometer-scale resolution for regional forecasting. They also use globally consistent reanalysis products such as the fifth-generation European Centre for Medium-Range Weather Forecasts reanalysis (ERA5), which provide high-quality estimates of the recent atmospheric state and large-scale drivers for local prediction. [5,6] However, solving these systems of coupled partial differential equations has high computational cost, which constrains spatial and temporal resolution and limits forecast throughput. Consequently, this creates a scalability barrier for continuous site-specific forecasting at wind farm level [7].

In contrast to computationally intensive physical methods, statistical methods provide efficient, data-driven predictions by modeling historical wind speed data. Early statistical methods include Autoregressive (AR) and Autoregressive Moving Average (ARMA) models, which capture short-range temporal dependencies in the recent wind history [8,9]. These models assume that the underlying time series is weakly stationary, with stable mean, variance and autocovariance over time. For non-stationary data, Autoregressive Integrated Moving Average (ARIMA) generalizes ARMA by applying differencing to remove trends. Seasonal ARIMA (SARIMA) further extends this structure with seasonal autoregressive and moving-average terms to capture strong diurnal or annual cycles. [10,11] Furthermore, to integrate external atmospheric drivers, Autoregressive Integrated Moving Average with Exogenous Variables (ARIMAX) and Seasonal Autoregressive Integrated Moving Average with Exogenous Variables (SARIMAX) extend these models with exogenous variables such as pressure as linear inputs [12,13]. Although statistical methods are computationally efficient and often competitive for very short-term forecasts at a single site, they rely on a linear superposition of effects. As a result, they struggle to represent multi-scale, regime-dependent, nonlinear flow structures. They also tend to lose accuracy at longer horizons and during fast-changing wind events, where machine learning and deep learning methods usually perform better.

To overcome the limited expressive capacity of stationary linear models, machine-learning methods enlarge the hypothesis class to nonlinear function approximators via activation-driven compositions, hierarchical partitioning and kernel-induced feature embeddings. Some of the recently used machine learning models for WSP include Support Vector Regression (SVR) [14,15] and Random Forest (RF) [16,17]. Empirical studies report that SVR often outperforms simpler methods such as back-propagation neural networks under variable wind conditions. Its kernel mapping captures nonlinear flow regimes, and its margin-based objective controls overfitting. In contrast, RF uses an ensemble of decorrelated decision trees to model interaction effects between meteorological drivers and provides stable forecasts across sites and horizons. [15–17] However, these shallow learners represent time through fixed lag windows or handcrafted temporal statistics rather than through an explicit dynamical state. This design forces manual feature selection, limits long-range temporal modeling, prevents correction of accumulated multi-step forecast error, and reduces stability under multi-scale wind variability. [18–20]

To address these issues, deep learning models treat wind speed forecasting as a sequential dynamical system. They learn the latent state directly from data instead of relying on manual feature selection or fixed lag windows. Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU) maintain an internal memory that propagates information through time. This memory allows the model to encode long-range temporal dependencies and adapt predictions as conditions evolve, rather than freezing the statistics learned during training. [21–23] The gated structure in LSTM and GRU learns which past information to retain and which to discard. This mechanism mitigates vanishing gradients and reduces multi-step rollout error because the network can update its hidden state over successive forecast horizons. [21–23] Temporal Convolutional Networks (TCNs) address multi-scale variability through dilated causal convolutions. These convolutions create a receptive field that spans multiple temporal scales, so short-term gusts and slow synoptic trends contribute at different dilation levels. [24,25] In addition, automated hyperparameter search and adaptive calibration reduce reliance on handcrafted feature engineering. Examples include genetic tuning of LSTM, C-LSTM mechanisms that adjust convolutional gates, and particle-swarm-guided feature selection for GRU. These methods can produce site-specific configurations that respond to local terrain and regime structure. [21,22,26,27] However, these recurrent and convolutional architectures still propagate information through either sequential hidden states or finite receptive fields. This makes it difficult to model very long-range global dependencies and joint spatio-temporal interactions, which motivates architectures that learn direct cross-time attention.

Transformer models address the remaining limitations of recurrent and convolutional networks by tokenizing the input sequence and applying global self-attention. These models can be further classified into three main categories, namely point-wise tokenization models, patch-based tokenization models and variate-wise tokenization models. Point-wise tokenization treats each time step as a token, as in the original Transformer and long-horizon forecasters such as Autoformer and FEDformer. These models decompose series structure into trend, seasonal, and frequency components to capture global periodicity. [28–30] Patch-based tokenization segments the history into short temporal patches and treats each patch as a token, enabling efficient long-context modeling in Crossformer [31] and Patch Time Series Transformer (PatchTST) [32]. Variate-wise tokenization inverts the view and treats each physical variable (for example wind speed, temperature, pressure) as a token so attention can learn cross-variable coupling, as in inverted Transformer (iTransformer) [33] and TimeXer [34]. These designs achieve strong accuracy on medium- and long-range meteorological and energy benchmarks by leveraging global structure and multivariate interactions [32,33]. However, at short-term WSP horizons, rapid gust ramps and terrain-driven transients dominate the target signal. Long-horizon Transformers bias toward smoothed trend and seasonal components and toward patch-level averages, which can attenuate high-frequency local dynamics and degrade near-term turbine-scale forecasts [35].

These limitations of deep learning models are partially addressed by hybrid models that exploit the complementary strengths of different architectures and methods. One common approach is decomposition-based forecasting. In this setting, methods such as Variational Mode Decomposition and Empirical Mode Decomposition first decompose the signal and then heuristically select the most informative intrinsic mode functions (IMFs) to improve WSP. [36,37] However, these methods are computationally expensive and still require manual feature selection. In addition, there is a growing line of convolutional–recurrent hybrids in which a Convolutional Neural Network (CNN) extracts spatial structure and an LSTM network models temporal dependencies. A shallow hybrid model couples a TCN with an LSTM module in parallel to improve WSP on edge devices. The study also introduces an adaptive optimization algorithm that outperforms the standard Tree-structured Parzen Estimator (TPE) by adjusting the exploration–exploitation threshold during search. [38] The proposed hybrid model outperforms several Transformer baselines, including iTransformer [33]. A related architecture in [39] also combines TCN with LSTM but first applies a Savitzky–Golay filter to denoise the input signal. This method consistently outperforms LSTM, TCN, Convolutional Long Short-Term Memory (ConvLSTM) and TCN–LSTM baselines in WSP. A further line of work applies 3D convolution to extract spatio-temporal features from multi-variable meteorological fields. It then uses a deep ConvLSTM module with ResNet-style skip connections to improve temporal modeling while mitigating vanishing-gradient effects. [40] This architecture is further extended in [3,41] to build physics-informed predictors for wind speed and temperature. In parallel, other studies introduce an explicit spatial representation to capture wind-flow structure. They then fuse this representation with a recurrent backbone to improve short-term WSP and move toward physically informed, site-aware forecasting. [42] Finally, authors in [43] proposed a physics-informed Temporal Convolutional Network-Temporal Fusion Transformer (TCN-TFT) model that integrates a TCN for multi-scale feature extraction with a Temporal Fusion Transformer (TFT) for temporal dynamics and employs quantile regression to generate predictive intervals. The model also enforces physical consistency by incorporating Bernoulli's principle and the ideal gas law, enhancing reliability without substantially increasing interval width. However, these advances still optimize global error over mixed regimes and only weakly encode atmospheric physics, which leaves the behavior during rapid wind-speed fluctuations underexplained.

1.2. Shortcomings in existing work

Most recent time series forecasting models for wind treat the atmosphere as a generic multivariate sequence and optimize a global Mean Squared Error (MSE) or Mean Absolute Error (MAE) objective over the entire history. Convolutional recurrent hybrids such as ConvLSTM, Convolutional Residual Spatial-Temporal Long Short-Term Memory (CoReSTL), Savitzky–Golay-filtered Temporal Convolutional Network-Long Short-Term Memory (SZ-TCN-LSTM), Convolutional Neural Network-Bidirectional Gated Recurrent Unit (CNN-BiGRU) and graph-based bidirectional recurrent network (GBRN), together with Transformer-based architectures such as iTransformer and PatchTST [32,33,39,40,44–46] learn a single stationary parameter set that must interpolate across both calm and highly volatile periods. The empirical risk is dominated by low- and moderate-volatility hours, so optimization mainly reduces error in typical conditions while assigning very little effective weight to rapid wind-speed changes. In distributional terms, the predictor approximates the central region of the conditional law of wind speed and leaves the tails weakly constrained. Even when the loss function reweights tail samples through focal or curriculum-style terms, a single shared parameter set still compromises across regimes. Stronger gradients from volatile hours can then reduce accuracy during calm periods. As a result, these models often maintain good accuracy in slowly varying regimes. However, they exhibit their largest absolute and relative errors when wind speed changes rapidly, which are the regimes that matter most for short-term dispatch and reserve scheduling.

Several recent WSP models introduce partial physical structure, yet still inherit this limitation. Physics-informed LSTM (PI-LSTM) first constructs a spatial prior of the surrounding flow and then applies a purely data-driven recurrent predictor without explicit dynamical constraints, so it remains effectively statistical [42]. Physics-Informed Temporal Convolutional Network-Temporal Fusion Transformer (PIN-TCN-TFT) regularises the network with constraints derived from the Bernoulli equation and the ideal gas law. This design assumes steady, inviscid, and locally homogeneous flow, which is computationally efficient but remains a biased surrogate for the atmospheric boundary layer. [43]. When training enforces agreement with both accurate wind speed observations (ground truth) and this simplified physics surrogate, the optimization problem becomes ill-conditioned because the surrogate and the measurements induce conflicting gradients in volatile regimes and the resulting physics residual can drive negative transfer. In parallel, SZ-TCN-LSTM and related hybrids rely on external denoising and decomposition stages such as Savitzky-Golay filtering, Variational Mode Decomposition, Empirical Mode Decomposition, and evolutionary feature selection. These stages introduce additional hyperparameters and brittle interfaces. [39] None of these approaches uses a physically grounded volatility estimate to define regimes, route recurrent updates through specialized parameter sets, or weight the loss by atmospheric state. As a result, predictive behavior under rapid wind-speed fluctuations remains weakly controlled. Together, these issues leave a clear gap in current WSP methods. An effective predictor should define regimes through a physically grounded volatility measure, learn regime-specialized spatio-temporal dynamics, and optimize a loss that reflects regime-dependent operational risk. Accordingly, the main objective of this work is to develop a physics-informed deep learning framework for short-term wind speed prediction that can explicitly adapt to different atmospheric volatility regimes. More specifically, this study aims to determine whether a physically grounded volatility proxy can improve regime identification, whether regime-specialized temporal modeling can reduce forecast error under rapidly changing wind conditions, and whether volatility-adaptive optimization can further improve predictive accuracy, especially during extreme events.

1.3. Contributions

To achieve this objective, this work proposes the Volatility-Regulated Temporal Evolutionary Network (VORTEX), which combines a physics-based volatility estimate with regime-aware temporal modeling and adaptive optimization for short-term wind speed prediction. The overall contributions of this work can be summarized as:

- A physics-based volatility proxy that quantifies atmospheric instability via pressure-gradient force discrepancy and provides an interpretable, causally grounded driver for regime-aware forecasting.
- A Membership-Gated Convolutional Recurrent Module that uses volatility-dependent gating to specialize different recurrent parameter sets for distinct wind regimes instead of enforcing a single shared dynamic.
- A Differentiable Evolutionary Mixture layer that performs adaptive feature selection and suppression inside the network, removing the need for external filtering or manual feature engineering.
- A Volatility-Adaptive Hybrid loss that assigns regime-weighted penalties across the forecast horizon to emphasize extreme, high-volatility events that drive grid risk.

The rest of the paper is organized as follows. Section 2 presents the proposed methodology. Section 3 reports results and comparisons. Section 4 concludes.

2. Proposed methodology

This work presents a framework for extreme volatile WSP that integrates physical principles with deep learning to model complex spatio-temporal patterns under rapidly shifting volatility regimes. Traditional methods, which treat wind prediction as either a purely physical simulation or a purely statistical model, fail to capture the dynamic relationship between atmospheric physics and prediction uncertainty. The framework, illustrated in Fig. 1, uses four interconnected components. The first is a Physics-Based Volatility Proxy that quantifies atmospheric instability via pressure gradient force discrepancy. This volatility proxy is then passed through a fuzzy membership assignment block that classifies each input example into a volatility regime. The third module

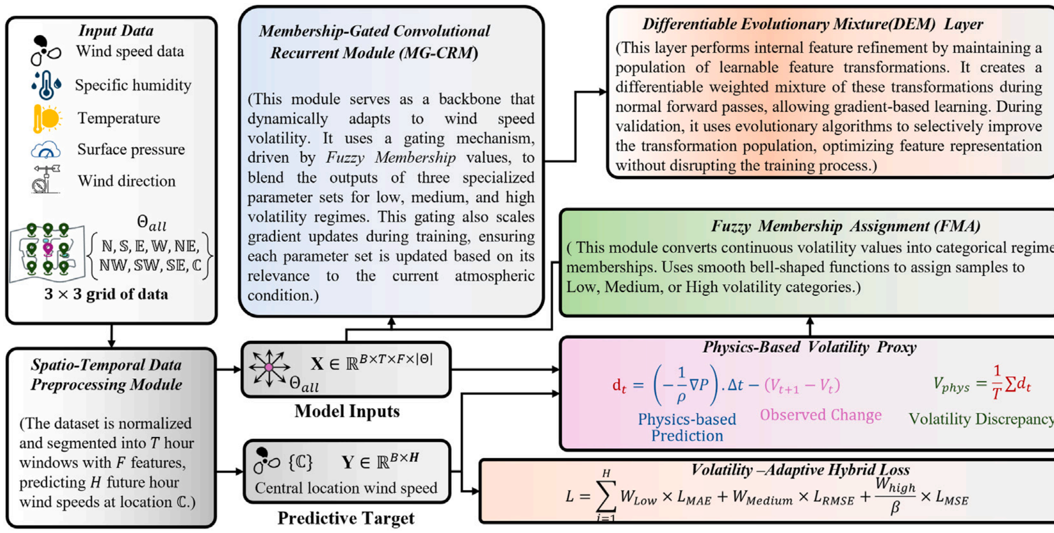


Fig. 1. Overall framework of the proposed extreme wind speed prediction pipeline. Meteorological inputs on a 3×3 grid are first used to compute the physics-based volatility proxy V_{phys}, which is then converted into regime memberships and used to guide the Volatility-Regulated Temporal Evolutionary Network (VORTEX) for the final wind speed forecast.

is an MG-CRM that maintains distinct sets of model weights for different volatility types. The output of the MG-CRM is passed through a Differentiable Evolutionary Mixture (DEM) layer, which refines the features and projects them for the final prediction. Together, the MG-CRM and DEM layers form a Volatility-Regulated Temporal Evolutionary Network (VORTEX), which adapts its computation to the measured volatility. The overall network is trained end-to-end with VAH-Loss, which adjusts the weights of different loss components based on the learned discrepancy for each example.

2.1. Physics-based volatility proxy

Conventional statistical measures of volatility, such as rolling standard deviation, quantify the magnitude of wind speed variations but lack physical interpretation. To provide a more foundational measure of atmospheric instability, we introduce a physics-based volatility proxy derived from the discrepancy between expected and observed wind acceleration. The primary driver of horizontal wind in the atmosphere is the Pressure Gradient Force (PGF). We leverage this fundamental relationship to establish a physical prior for expected wind behavior. The acceleration due to PGF is given by,

$$\mathbf{a}_{\text{pgf}} = -\frac{1}{\rho}\nabla p, \tag{1}$$

where ρ is the air density and ∇p is the pressure gradient. We compute the components of the pressure gradient using a centered difference scheme on the 3×3 grid surrounding the target location,

$$\frac{\partial p}{\partial x} \approx \frac{p_{\text{east}} - p_{\text{west}}}{2\Delta x}, \tag{2}$$

$$\frac{\partial p}{\partial y} \approx \frac{p_{\text{north}} - p_{\text{south}}}{2\Delta y}, \tag{3}$$

where $\Delta x = \Delta y = 100$ km is the horizontal grid spacing and pressure values are converted from the original kilopascals to pascals (1 kPa = 1000 Pa) to maintain dimensional consistency. The magnitude of the PGF-induced acceleration is given as:

$$|\mathbf{a}_{\text{pgf}}| = \sqrt{\left(-\frac{1}{\rho}\frac{\partial p}{\partial x}\right)^2 + \left(-\frac{1}{\rho}\frac{\partial p}{\partial y}\right)^2}. \tag{4}$$

Because the meteorological fields and wind-speed observations are available at hourly resolution, this work uses a first-order finite-interval approximation over $\Delta t = 3600$ s to translate the pressure-gradient-force magnitude into an expected hourly wind-speed tendency. The resulting $\Delta v_{\text{predicted}}$ is therefore not intended to resolve sub-hour turbulence or the complete atmospheric momentum budget. Instead, it serves as a coarse-resolution reference change under resolved large-scale pressure-gradient forcing, against which the observed hourly wind-speed change is compared. The predicted change in wind speed is then written as:

$$\Delta v_{\text{predicted}} = |\mathbf{a}_{\text{pgf}}| \cdot \Delta t. \tag{5}$$

This predicted change represents the expected wind speed variation if the pressure gradient force were the sole governing dynamic. We compare this physics-based expectation to the observed change in wind speed,

$$\Delta v_{\text{actual}} = v_{t+1} - v_t, \tag{6}$$

where v_t and v_{t+1} are the observed wind speeds at the central location at consecutive hours. The core of the volatility measure is the absolute discrepancy between these values and is given as:

$$d_t = |\Delta v_{\text{predicted}} - \Delta v_{\text{actual}}|. \tag{7}$$

This value provides a direct measure of mechanical forcing imbalance at the hourly scale. A high d_t indicates that the observed hourly wind-speed change departs from what is expected from the resolved large-scale pressure gradient alone. In this sense, d_t should be interpreted as a coarse-resolution forcing-mismatch indicator rather than as a full physical acceleration model. Unresolved processes, including turbulence, convection, local topographic effects, and sub-hour variability, are not explicitly modeled; instead, their aggregate influence contributes to the mismatch term. Hence d_t serves as an interpretable physics-based indicator of volatility that links more directly to the atmospheric state than purely statistical features. For each input sample, comprising T consecutive hours, this work computes a single volatility proxy value by averaging the instantaneous discrepancies as,

$$V_{\text{phys}} = \frac{1}{T}\sum_{t=1}^{T} d_t. \tag{8}$$

This V_{phys} serves as our physics-based volatility measure, which is provided as an input feature to the VORTEX architecture. Unlike a rolling standard deviation, V_{phys} flags hours when observed wind-speed changes contradict the pressure-gradient prediction.

2.2. Fuzzy volatility membership assignment

The V_{phys} values are fuzzified to quantify membership in volatility regimes using parameterized functions. This work employs generalized bell membership functions for their smooth transitions and ability to capture central tendency and distribution width, given as:

$$\mu_k(V_{\text{phys}}) = \frac{1}{1 + \left|\frac{V_{\text{phys}} - \gamma_k}{\alpha_k}\right|^{2\beta_k}}, \quad k \in \{\text{low}, \text{medium}, \text{high}\}, \tag{9}$$

where α_k controls the width, β_k controls the slope and γ_k determines the center of the membership function for each volatility regime. In this work, the fuzzy membership design is selected through the same TPE-based Optuna hyperparameter search used for the overall model configuration. The search evaluates the membership-function family, the normalization mode, and the scaling of the regime-definition parameters according to validation Root Mean Squared Error (RMSE). The selected configuration corresponds to the generalized-bell form of $\mu_k(V_{\text{phys}})$ with softmax normalization, unit scaling on both γ_k and α_k , and $\beta_k = 2.0$. After this design is selected, the final regime centers are instantiated from the training-set distribution of the volatility proxy, with $\gamma_{\text{low}} = \mu - \sigma$, $\gamma_{\text{medium}} = \mu$, and $\gamma_{\text{high}} = \mu + \sigma$, while the width is set as $\alpha_k = \sigma$ for all regimes. The same parameterization is then used to compute memberships for the training, validation, and test samples.

The sensitivity analysis further shows that changing the membership design away from this selected configuration does not improve validation accuracy, as reported later in Fig. 4. In particular, the generalized-bell form, softmax normalization, unit scaling on both γ_k and α_k, and $\beta_k = 2.0$ yield the lowest validation RMSE among the tested alternatives, which confirms that this configuration is the optimal membership choice for the final model.

2.3. Volatility-regulated temporal evolutionary network (VORTEX)

The VORTEX architecture processes spatio-temporal wind patterns while adapting to volatility regimes identified by the physics-based volatility proxy. The model takes the spatio-temporal input $\mathbf{X} \in \mathbb{R}^{B\times T\times F\times 3\times 3}$ containing F meteorological features across a 3×3 grid region centred on the prediction target. The overall design passes input from a 3D convolution layer to generate feature embeddings that capture local spatio-temporal patterns. These embeddings then pass through a cascade of Membership-Gated Convolutional Recurrent Modules (MG-CRM) (Section 2.3.1), with a residual connection after every two layers to avoid vanishing gradients. The features are then processed through a Differentiable Evolutionary Mixture (DEM) layer (Section 2.3.2) for feature refinement. The overall architecture of the proposed VORTEX network is illustrated in Fig. 2.

2.3.1. Membership-gated convolutional recurrent module (MG-CRM)

Traditional wind prediction architectures employ uniform computational strategies that fail to adapt to rapidly changing atmospheric volatility regimes. This limitation becomes critical during extreme wind events where static models either underfit volatile patterns or oversmooth calm conditions. To address this, we propose a Membership-Gated Convolutional Recurrent Module that dynamically routes computation through specialized parameter sets based on measured volatility levels.

The MG-CRM employs a hybrid design that balances specialization with parameter efficiency. While standard ConvLSTM uses a single recurrent kernel $\mathbf{W}_{\text{rec}}$ for all inputs, our approach maintains $K = 3$ distinct recurrent parameter sets $\{\mathbf{W}_{\text{rec},k}, \mathbf{b}_k\}_{k=1}^{3}$ specialized for low, medium and high volatility regimes, while keeping the input kernel $\mathbf{W}_{\text{input}}$ and gating mechanisms shared across all regimes.

Volatility memberships $m_k(t) \in [0, 1]$ for $k = 1, 2, 3$ are derived from the physics-based proxy V_{phys} through fuzzy membership functions and normalized such that $\sum_{k=1}^{3} m_k(t) = 1$. These memberships are computed per sample and remain constant across all time steps t within each sequence, with spatial broadcasting across height and width dimensions. This design ensures sample-specific routing without temporal leakage, as V_{phys} computes discrepancies over the input window $[t - L, t]$ using only contemporaneous and historical pressure and wind observations.

This design choice is motivated by three key considerations. First, input representation consistency ensures that raw meteorological observations in $\mathbf{x}_t$ are processed uniformly, providing stable feature extraction regardless of volatility regime. Second, temporal dynamics specialization focuses volatility adaptation on the hidden state $\mathbf{h}_{t-1}$, which encodes accumulated temporal patterns that benefit most from regime-specific processing. Third, parameter efficiency is achieved by specializing only the recurrent connections, which typically dominate ConvLSTM parameter counts. For input channels C_{in}, filters F and kernel size $k \times k$, the

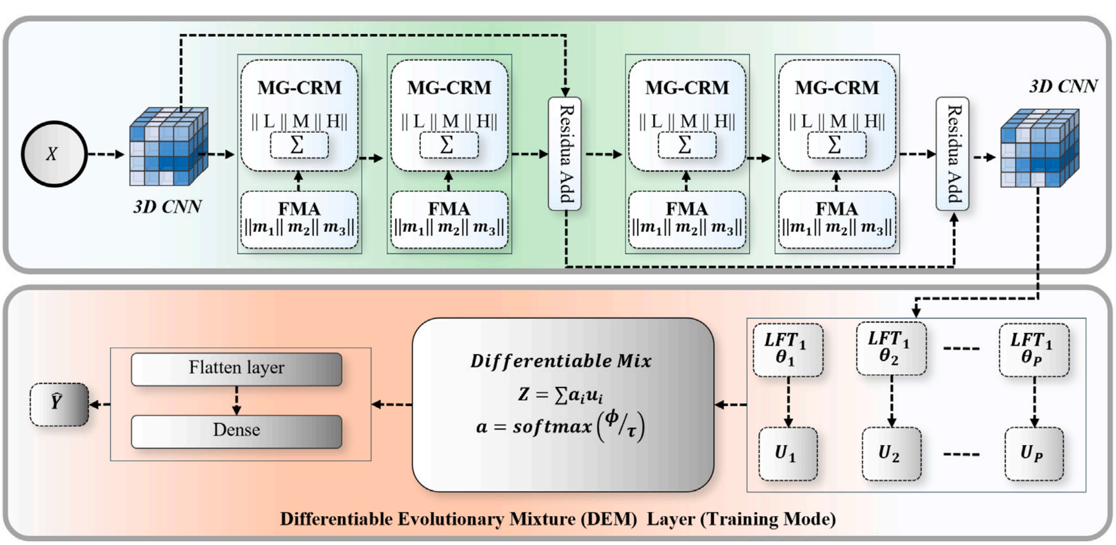


Fig. 2. Detailed architecture of VORTEX. A 3D convolutional embedding feeds a cascade of Membership-Gated Convolutional Recurrent Modules with residual connections after every two layers, followed by a Differentiable Evolutionary Mixture layer that refines features before the final wind speed prediction. The physics-based volatility proxy provides regime-aware guidance for adaptive routing.

parameter ratio is:

$$\frac{N_{\text{rec}}}{N_{\text{in}}} = \frac{4F^2k^2}{4FC_{\text{in}}k^2} = \frac{F}{C_{\text{in}}} \tag{10}$$

With $F \gg C_{\text{in}}$ in deep networks, specializing recurrent weights ($N_{\text{rec}} = 4F^2k^2$) provides maximum adaptation impact while controlling complexity. The forward pass implements this hybrid specialization through parallel recurrent convolutions combined via membership-weighted summation. The convolution operator C uses same padding with kernel size $k \times k$, unit stride, no dilation and produces $4F$ output channels:

$$\mathbf{z}_{\text{input}}(t) = C(\mathbf{x}_t, \mathbf{W}_{\text{input}}) \in \mathbb{R}^{B\times 4F\times H\times W}, \tag{11}$$

$$\mathbf{z}_{\text{rec},k}(t) = C(\mathbf{h}_{t-1}, \mathbf{W}_{\text{rec},k}) + \mathbf{b}_k \quad \text{for } k = 1, 2, 3, \tag{12}$$

$$\mathbf{z}_{\text{rec}}(t) = \sum_{k=1}^{3} m_k(t)\cdot \mathbf{z}_{\text{rec},k}(t), \tag{13}$$

$$\mathbf{z}(t) = \mathbf{z}_{\text{input}}(t) + \mathbf{z}_{\text{rec}}(t). \tag{14}$$

The combined signal $\mathbf{z}(t)$ is then split into four contiguous gate segments of size F channels each:

$$i(t) = \sigma(\mathbf{z}(t)[\dots, : F]), \quad f(t) = \sigma(\mathbf{z}(t)[\dots, F : 2F]), \tag{15}$$

$$c_c(t) = \tanh(\mathbf{z}(t)[\dots, 2F : 3F]), \quad o(t) = \sigma(\mathbf{z}(t)[\dots, 3F : 4F]), \tag{16}$$

$$c_t = f(t)\odot c_{t-1} + i(t)\odot c_c(t), \quad h_t = o(t)\odot \tanh(c_t). \tag{17}$$

This architecture ensures that while recurrent transformations are regime-specific, the gating mechanisms remain adaptive to the combined input and specialized hidden state, providing a balanced approach to volatility-aware temporal modeling. During backpropagation, gradients flow through fixed membership values (stopped gradient) to prevent degenerate routing, with updates accumulated over time:

$$\frac{\partial\mathcal{L}}{\partial\mathbf{W}_{\text{rec},k}} = \sum_t m_k(t)\cdot\frac{\partial\mathcal{L}}{\partial\mathbf{z}_{\text{rec}}(t)}\cdot\frac{\partial\mathbf{z}_{\text{rec},k}(t)}{\partial\mathbf{W}_{\text{rec},k}}, \tag{18}$$

$$\frac{\partial\mathcal{L}}{\partial\mathbf{b}_k} = \sum_t m_k(t)\cdot\frac{\partial\mathcal{L}}{\partial\mathbf{z}_{\text{rec}}(t)}. \tag{19}$$

This design enables volatility-aware temporal modeling while maintaining computational efficiency, with floating-point operations (FLOPs) and memory scaling linearly with K. The membership normalization and gradient stopping ensure training stability without additional regularization and the sample-level routing provides consistent volatility adaptation throughout each input sequence while preventing information leakage from future time steps.

2.3.2. Differentiable evolutionary mixture (DEM) layer

Feature maps from earlier layers often carry redundant or weak signals that increase prediction error. Many recent methods insert a separate feature selection stage that uses evolutionary search and Pearson correlation on inputs. That choice adds complexity and extra hyperparameters. The DEM layer performs feature refinement inside the model. It learns mixture weights during normal training and runs evolution on validation data only, so gradients flow without interruption while the population of transformations still improves. The layer maintains a population of P Learnable Feature Transformations (LFTs). Each LFT is a convolutional autoencoder that operates on the incoming features $\mathbf{F} \in \mathbb{R}^{B\times T\times F\times\Theta}$ and produces

$$\mathbf{U}_i = \text{LFT}_i(\mathbf{F}, \boldsymbol{\theta}_i) \in \mathbb{R}^{B\times T\times F\times\Theta}, \quad i = 1, \dots, P. \tag{20}$$

Where $\mathbf{U}_i$ is the transformed feature tensor from the ith LFT and $\boldsymbol{\theta}_i$ is its parameter set. The forward path forms a differentiable global mixture with trainable logits $\boldsymbol{\phi} \in \mathbb{R}^P$ and temperature $\tau > 0$

Algorithm 1 DEMLayer forward and validation evolution.

procedure FORWARD($\mathbf{F}$)
  $\mathbf{U}_i \leftarrow \text{LFT}_i(\mathbf{F}, \boldsymbol{\theta}_i)$ for $i = 1, \dots, P$
  $\alpha \leftarrow \text{softmax}\big((\boldsymbol{\phi} - \max(\boldsymbol{\phi}))/\tau\big)$
  $\mathbf{Z}_{\text{dem}} \leftarrow \sum_{i=1}^{P} \alpha_i\,\mathbf{U}_i$
  return $\mathbf{Z}_{\text{dem}}$
end procedure
procedure EVOLUTIONARYUPDATE($\{\mathbf{F}_n\}$)
  for $i = 1$ to P **do**
    $\mathcal{L}^{(i)}_{\text{recon}} \leftarrow \frac{1}{N_{\text{val}}}\sum_n \|\mathbf{F}_n - \text{LFT}_i(\mathbf{F}_n, \boldsymbol{\theta}_i)\|_2^2$
    $\text{fitness}_i \leftarrow 1/(\mathcal{L}^{(i)}_{\text{recon}} + \epsilon)$
  end for
  elites $\leftarrow$ top_k(fitness)
  for $m = 1$ to $P - k$ **do**
    parents $\leftarrow$ select(elites)
    child $\leftarrow$ crossover(parents)
    child $\leftarrow$ mutate(child)
    replace lowest fitness LFT with child
  end for
end procedure

$$\alpha_i = \frac{\exp(\phi_i/\tau)}{\sum_{j=1}^{P}\exp(\phi_j/\tau)}, \quad \sum_{i=1}^{P}\alpha_i = 1,\ \alpha_i \in [0,1],\ \mathbf{Z}_{\text{dem}} = \sum_{i=1}^{P}\alpha_i\,\mathbf{U}_i \in \mathbb{R}^{B\times T\times F\times\Theta}. \tag{21}$$

Where α_i is the mixture weight for the ith transformation, $\boldsymbol{\phi}$ is the vector of trainable logits, τ controls softness of the weights, $\mathbf{Z}_{\text{dem}}$ is the DEM output and i indexes the P transformations. A stable form centers the logits before softmax

$$\alpha = \text{softmax}\big((\boldsymbol{\phi} - \max_j \phi_j)/\tau\big).$$

During task training the model minimizes the prediction loss and updates $\boldsymbol{\phi}$ and $\{\theta_i\}_{i=1}^{P}$. During validation the model improves the LFT population without gradients. It computes a reconstruction objective and a fitness score for each LFT

$$\mathcal{L}^{(i)}_{\text{recon}} = \frac{1}{N_{\text{val}}}\sum_{n=1}^{N_{\text{val}}}\left\|\mathbf{F}_n - \text{LFT}_i\big(\mathbf{F}_n, \theta_i\big)\right\|_2^2, \qquad \text{fitness}_i = \frac{1}{\mathcal{L}^{(i)}_{\text{recon}} + \epsilon}. \tag{22}$$

Where N_{val} is the number of validation batches, $\mathbf{F}_n$ is the nth validation input, ϵ avoids division by zero and fitness_i ranks LFT i. The procedure then selects elites, applies crossover and mutation to create children and replaces the lowest fitness members. This process refines useful transformations and reduces redundant ones while the training graph stays simple and fully differentiable.

2.4. Volatility-adaptive hybrid loss

Static loss functions apply a uniform penalty to prediction errors, which creates a fundamental limitation for wind prediction where the cost of error varies significantly with atmospheric stability. This work introduces a volatility-adaptive hybrid loss that dynamically recalibrates its objective function in response to the identified wind regime. The complete loss function combines multiple error metrics, each weighted by a physics-informed membership value and is given as follows:

$$\mathcal{L} = \sum_{i=1}^{H}\left[w_{\text{low}}\cdot\mathcal{L}^{(i)}_{\text{MAE}} + w_{\text{medium}}\cdot\mathcal{L}^{(i)}_{\text{RMSE}} + \frac{w_{\text{high}}}{\beta}\cdot\mathcal{L}^{(i)}_{\text{MSE}}\right], \tag{23}$$

where $\beta = 1$ m s^{-1} ensures units consistency, H is the prediction horizon count, w represents the volatility-adaptive weights and $\mathcal{L}$ denotes the constituent loss terms. The raw membership values m_k undergo normalization to ensure stable optimization:

$$\tilde{m}_k = \frac{m_k}{\sum_{j=1}^{3} m_j + \epsilon}, \tag{24}$$

where $\tilde{m}_k$ represents the normalized membership and ϵ provides numerical stability. A focus factor amplifies the loss sensitivity during transitional conditions between volatility regimes:

$$f = 1 + (\tilde{m}_{\text{high}} - \tilde{m}_{\text{low}}). \tag{25}$$

This focus factor modulates the final weight assignments through:

$$w_{\text{low}} = \tilde{m}_{\text{low}} \cdot \frac{1}{\max(f, 0.5)}, \tag{26}$$

$$w_{\text{medium}} = \tilde{m}_{\text{medium}}, \tag{27}$$

$$w_{\text{high}} = \tilde{m}_{\text{high}} \cdot f. \tag{28}$$

The complete system employs standard error metrics tailored to each volatility regime:

$$\mathcal{L}^{(i)}_{\text{MAE}} = \frac{1}{N} \sum_{i=1}^{N} |y_i - \hat{y}_i|, \tag{29}$$

$$\mathcal{L}^{(i)}_{\text{RMSE}} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2}, \tag{30}$$

$$\mathcal{L}^{(i)}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2. \tag{31}$$

where N is the total number of examples. This formulation is designed to leverage the intrinsic properties of each error metric: MAE during stable conditions (low volatility) for its prediction accuracy in small fluctuations, RMSE for balanced sensitivity during moderate winds and MSE during high volatility to aggressively penalize large errors common in extreme wind events. The weighting mechanism ensures mathematical consistency while aligning optimization priorities with physically measured volatility characteristics.

3. Results and comparison

This section evaluates VORTEX across multiple geographically diverse datasets and compares its performance against state-of-the-art models for extreme wind speed prediction.

3.1. Dataset description and preprocessing

This work uses four National Aeronautics and Space Administration Prediction Of Worldwide Energy Resources (NASA POWER) Source Native Resolution Hourly datasets, accessed on 2025/03/21, featuring hourly wind speed measurements from Dodge City (37.75°N, 100.01°W) with data from 2009 to 2024, Sarmiento (45.60°S, 69.08°W) with data from 2010 to 2024, Jiuquan (40.50°N, 96.90°E) with data from 2008 to 2024, and Pincher Creek (49.55°N, 114.05°W) with data from 2014 to 2024. For each site, hourly records are retrieved in Local Solar Time (LST) for the central grid point and its eight surrounding grid points to form a 3×3 spatial layout, using wind direction at 50 m (WD50M), wind speed at 50 m (WS50M), surface pressure (PS), temperature at 2 m (T2M), and specific humidity at 2 m (QV2M) as input variables. The source defines the missing-data code as -999, but no missing values are present in the retrieved records used in this study. The hourly records from the nine grid points are aligned by timestamp, organized into a common 3×3 spatial tensor, and then converted into model samples using a 24-hour input window, a 6-hour stride, and a subsequent 6-hour prediction horizon at the central grid point. The resulting chronological training, validation, and test splits for each site are reported in Table 1. Finally, mean-variance normalization is applied using statistics estimated from the training set only and then applied unchanged to the validation and test sets.

Table 1
Chronological dataset split for each location.

City	Total Hours	Training Hours	Validation Hours	Test Hours
Dodge City	140,256	112,205	14,026	14,025
Sarmiento	131,496	105,197	13,150	13,150
Jiuquan	149,064	119,252	14,907	14,905
Pincher Creek	96,384	77,108	9639	9637

Table 2
Statistical summary of wind speed (m s^{-1}) at the central locations of the four study sites.

City	Minimum	Maximum	Mean	Variance
Dodge City	0.03	30.49	7.37	10.56
Sarmiento	0.06	31.54	9.13	16.27
Jiuquan	0.02	23.89	6.59	12.07
Pincher Creek	0.01	21.55	5.76	8.45

The datasets show diverse statistical properties as in Table 2. Sarmiento has the highest variance, indicating volatile wind conditions, while Pincher Creek has the lowest mean and variance, suggesting calmer winds. The range between minimum and maximum values confirms extreme wind events across all locations. Fig. 3 illustrates distinct seasonal patterns in monthly mean variance. Dodge City and Jiuquan show peak variance in April, with lows in August and July, respectively. Pincher Creek has maximum variance in February and minimum in August, while Sarmiento peaks in November and reaches its minimum in July. These cyclical patterns provide a diverse test setting across sites and seasons.

3.2. Experimental setup

This study establishes a standardized experimental framework for WSP using spatio-temporal data. Each dataset undergoes preprocessing to generate input sequences of 24 hourly observations across a 3×3 spatial grid, with each grid point containing five meteorological features (temperature, humidity, wind speed, wind direction and pressure). The prediction task estimates wind speed at the central grid point for a 6-hour horizon using the preceding 24 hours of data. A sliding window approach with a 6-hour stride generates training samples while maintaining temporal relationships. All models are optimized using TPE. Baseline models and other state-of-the-art (SOTA) comparators employ MSE loss, while our proposed VORTEX utilizes the volatility-adaptive regime loss (VAH-Loss). Training and testing of all models are performed using an NVIDIA RTX 3080 Ti graphics processing unit (GPU) (Table 3).

3.3. Evaluation metrics

We evaluate the proposed model with three metrics that capture complementary aspects of accuracy. The first is MAE, given as follows,

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^{N} \left| y_i - \hat{y}_i \right|, \tag{32}$$

where y_i is the observed wind speed in m s^{-1}, $\hat{y}_i$ is the predicted wind speed in m s^{-1}, and N is the number of test samples. MAE summarizes the typical deviation between predictions and observations. Besides MAE, this work also compares models' performance on the basis of RMSE, given as follows,

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} \left(y_i - \hat{y}_i \right)^2}, \tag{33}$$

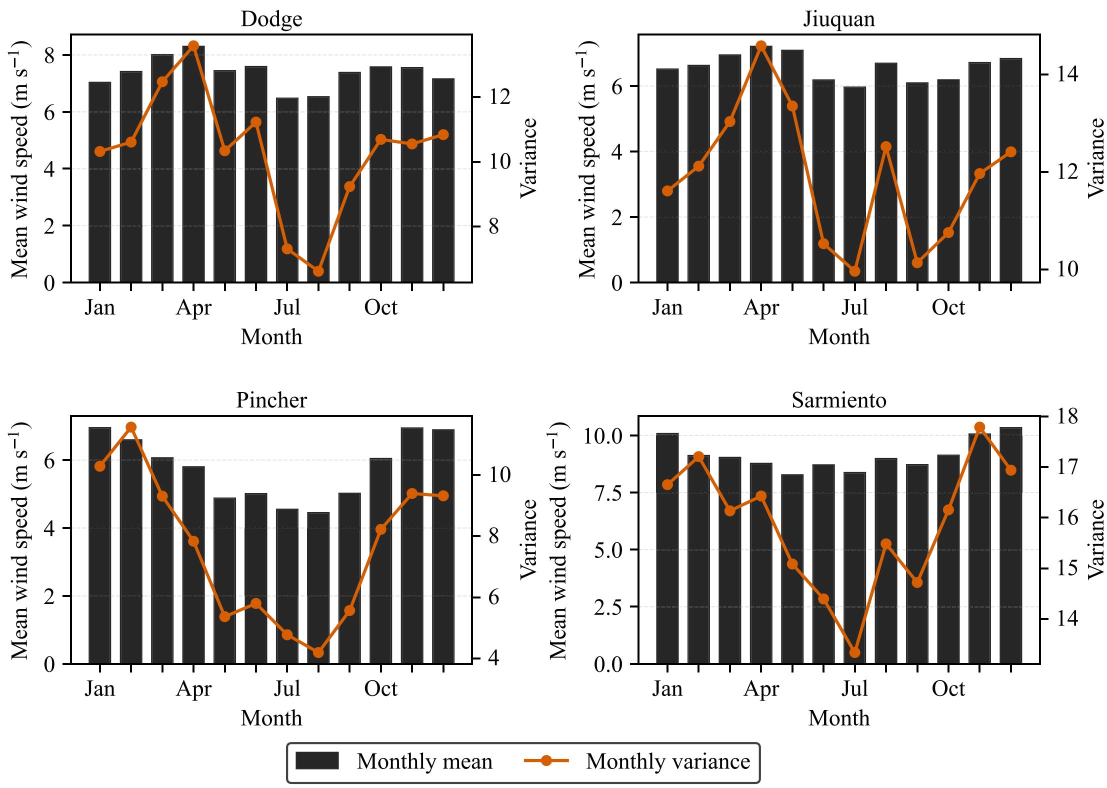


Fig. 3. Monthly mean variance of wind speed for the four datasets.

Table 3
Experimental configuration parameters.

Parameter	Value
Maximum epochs	1000
Early stopping patience	10 epochs
Optimization algorithm	TPE
Learning rate	1e-4
Batch size	32
SOTA loss function	MSE
Proposed loss function	VAH-Loss
Input sequence length	24 h
Spatial grid	3×3
Features per point	5
Temporal stride	6 h
Prediction horizon (H)	6 h
Training/Testing hardware	NVIDIA RTX 3080 Ti

RMSE places greater weight on large errors and often reacts to gusts and ramps. Finally, this work also reports the coefficient of determination (R^2),

$$R^2 = 1 - \frac{\sum_{i=1}^{N}\left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{N}\left(y_i - \bar{y}\right)^2}. \tag{34}$$

R^2 is dimensionless, can be negative when the model underperforms the mean predictor and moves toward 1 as fit improves. Together, these metrics provide a balanced view of typical error, tail sensitivity and explained variance. We report all results per site and per forecast horizon at the native sampling interval.

3.4. Hyperparameter sensitivity analysis

This work evaluates hyperparameter sensitivity through a short-run validation protocol used to identify a stable VORTEX configuration before full training. Each candidate setting is trained for 10 epochs and then evaluated on the validation set using RMSE. After selecting the best value of each hyperparameter, this work perturbs one setting at a time while fixing all remaining settings at their selected values. The resulting analysis therefore measures local sensitivity around the selected configuration under a common and computationally efficient search budget. For the loss-design study, this work additionally compares five alternative training objectives under the same 10-epoch protocol, namely the selected Volatility-Adaptive Hybrid loss (VAH), a pure MSE loss, a Hybrid without Focus weighting (HWF), a Uniform Weights variant (UW), and a High-Regime RMSE variant (HR-RMSE), where the high-volatility term uses RMSE instead of the selected MSE-based penalty. The reported values in this subsection are intermediate validation results used for search-space exploration rather than final test results.

Fig. 4 summarizes the local sensitivity pattern in terms of ΔRMSE relative to the selected setting. The lowest validation RMSE is obtained for the generalized-bell form of $\mu_k(V_{\text{phys}})$, softmax normalization of m_k, unit scaling on both γ_k and α_k, $\beta_k = 2.00$, the selected VAH loss, and $K = 3$ volatility regimes. The regime-count analysis shows that performance is sensitive to this choice but remains well behaved. When the number of regimes is reduced from $K = 3$ to $K = 2$, the validation RMSE increases from 1.98 to 2.35 m/s, which suggests that a two-regime partition is too coarse to separate low-, medium-, and high-volatility behavior. When the number of regimes is increased to $K = 4$, the RMSE rises to 2.13 m/s, which suggests that further partitioning weakens the effective sample support available to each specialized pathway without providing a compensating gain in expressiveness. Together, these results indicate that $K = 3$ provides the best balance between regime resolution and parameter specialization.

For the loss variants, VAH yields the lowest RMSE of 1.98 m/s, whereas MSE, HWF, UW, and HR-RMSE increase the validation RMSE to 2.50, 2.21, 2.15, and 2.91 m/s, respectively. For the network architecture, the selected setting corresponds to $F_1 = 16$, $F_2 = 32$, and $k = (1, 1)$, together with ϕ_c = `selu` and ϕ_r = `softsign`. The strongest sensitivity

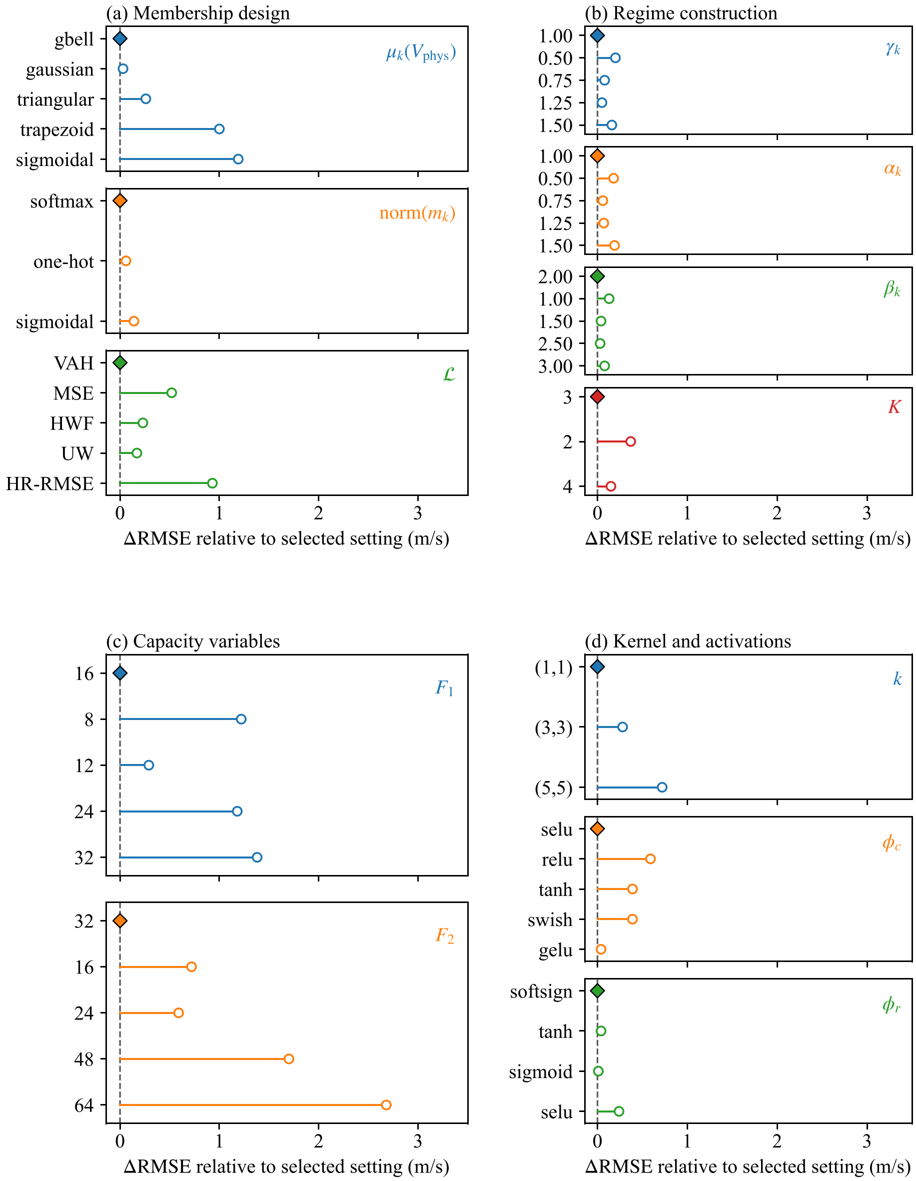


Fig. 4. Hyperparameter sensitivity analysis based on 10-epoch validation runs. The horizontal axis shows ΔRMSE relative to the selected configuration, where 0 corresponds to the lowest validation RMSE of 1.98 m/s. Panel (a) reports membership-design variables, panel (b) reports regime-construction variables, panel (c) reports capacity variables, and panel (d) reports kernel and activation variables.

appears in the capacity variables, especially F_2, where changing the second-stage filters to 16, 24, 48, and 64 increases the RMSE to 2.70, 2.57, 3.68, and 4.66 m/s, respectively. The remaining variables also show consistent degradation away from the selected configuration, including $F_1 = 8, 12, 24, 32$ with RMSE values of 3.20, 2.27, 3.16, and 3.36 m/s, $k = (3, 3)$ and $(5, 5)$ with RMSE values of 2.26 and 2.70 m/s, ϕ_c = `relu`, `tanh`, `swish`, and `gelu` with RMSE values of 2.57, 2.37, 2.37, and 2.02 m/s, and ϕ_r = `tanh`, `sigmoid`, and `selu` with RMSE values of 2.02, 1.99, and 2.22 m/s.

To complement the local hyperparameter analysis, this work further evaluates the seed-to-seed stability of the final selected configuration. Fig. 5 shows the validation RMSE over five random seeds. The resulting values remain tightly clustered, with mean of 2.03 m/s, standard deviation of 0.08 m/s, and range of 1.91–2.11 m/s, which indicates that the selected VORTEX configuration remains practically stable despite the stochastic evolutionary update.

3.5. Comparisons against baseline models

This work compares VORTEX with nine state-of-the-art (SOTA) time-series prediction models that span complementary paradigms. Two Transformer-based general-purpose models, iTransformer [33] and PatchTST [32], serve as attention-centric baselines. ConvLSTM [44] serves as a spatio-temporal recurrent baseline closely related to our design and CoReSTL [40] augments ConvLSTM with skip connections. We also include two physics-informed wind-speed-prediction models, PIN-TCN-TFT [43] and a spatial-prior fusion framework, PI-LSTM [42]. In addition, we evaluate four recent convolutional recurrent hybrids for short-term wind speed prediction: SZ-TCN-LSTM [39], CNN-BiGRU [45] and the spatiotemporal correlation method GBRN [46]. Together, these baselines cover attention, recurrent, physics-informed and hybrid designs and provide a broad and domain-relevant basis for comparison across sites and horizons. We select these baselines because they

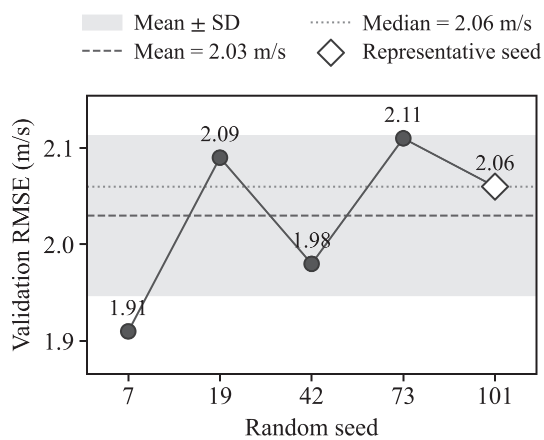


Fig. 5. Validation RMSE of the selected VORTEX configuration over five random seeds. The dashed line denotes the mean RMSE, the dotted line denotes the median RMSE, and the shaded band denotes mean ± one standard deviation. The highlighted diamond marks the representative seed, chosen as the run closest to the multi-seed median.

represent strong published architectures for short-term WSP across the main modeling families. Therefore, any improvement by VORTEX is more likely to reflect the value of volatility-aware, multi-gated physics integration than the weakness of the baselines. The experimental results demonstrate that VORTEX consistently outperforms state-of-the-art baselines across all four geographically diverse datasets. Table 4 presents the comprehensive comparison of average performance over the 6-hour prediction horizon.

On the Dodge City dataset, VORTEX achieves the strongest performance against CoReSTL [40], a ConvLSTM with skip connections that represents the most competitive baseline. The proposed model reduces RMSE by 4.33% from 1.64 to 1.57 m s^{-1} and decreases MAE by 5.56% from 1.19 to 1.12 m s^{-1}. The R^2 improvement reaches 4.26%, increasing from 0.71 to 0.74. These gains demonstrate VORTEX's effectiveness in handling the moderate volatility regime characteristic of this location. On the other hand, for the Jiuquan dataset, VORTEX exhibits particularly strong performance improvements. Against the strongest baseline PI-LSTM [3], VORTEX reduces RMSE by 7.64% from 1.61 to 1.48 m s^{-1} and MAE by 6.75% from 1.13 to 1.06 m s^{-1}. The R^2 value improves by 4.12% from 0.78 to 0.81. This enhancement suggests the framework's volatility-adaptive mechanisms effectively capture the complex wind patterns in this region.

Moving toward the Pincher Creek dataset results show the most pronounced improvements, with VORTEX outperforming CoReSTL by 9.23% in RMSE (1.13 to 1.03 m s^{-1}) and 10.28% in MAE (0.83 to 0.75 m s^{-1}). The R^2 increases by 3.50% from 0.83 to 0.86. These results indicate the model's particular strength in lower-variance wind regimes where precise prediction of subtle variations becomes critical. Finally, on the Sarmiento dataset, which exhibits the highest wind speed variance, VORTEX maintains consistent superiority over PI-LSTM. The model achieves a 7.34% RMSE reduction (1.34 to 1.25 m s^{-1}) and 4.59% MAE improvement (0.94 to 0.89 m s^{-1}). The R^2 increases by 1.61% from 0.90 to 0.91, demonstrating accurate performance even in highly volatile conditions where many models struggle.

There is a pattern in the comparative results across sites. In low-variance regimes, such as Pincher Creek with variance 8.45 m^2 s^{-2} in Table 2, CoReSTL is the strongest baseline in Table 4. Its shared convolutional kernels and recurrent state capture short-range spatial coupling and persistent temporal dynamics without auxiliary constraints. In contrast, in high-variance regimes such as Sarmiento with 16.27 m^2 s^{-2}, PI-LSTM uses spatial-prior fusion to compress the 3×3 field into a compact embedding. This compression suppresses peripheral noise and yields cleaner inputs, although it can discard fine-scale structure that remains useful under steadier flow. VORTEX advances beyond both by estimating instability with a pressure-gradient-force discrepancy and by routing updates through membership-gated recurrent parameters. This regime conditioning emphasizes local spatial detail when variance is low and prioritizes volatility-aware computation when variance is high, rather than using one stationary kernel set for all conditions.

Fig. 6 presents the temporal evolution of prediction accuracy across all horizons. VORTEX consistently maintains superior performance throughout the entire 6-hour prediction window. The stability of these improvements across increasing lead times demonstrates the model's accuracy against error accumulation, a common challenge in multi-step forecasting. The volatility-adaptive loss function and regime-specific parameterization effectively mitigate performance degradation over longer horizons.

The statistical significance analysis in Fig. 7 provides rigorous validation of VORTEX's performance advantages. The Wilcoxon signed-rank test results demonstrate that the improvements are statistically significant ($p < 0.01$) across all metrics and datasets. This statistical evidence indicates that VORTEX's superior performance does not occur by chance but stems from the fundamental advantages of its physics-informed volatility adaptation and specialized architectural components. The consistency across diverse geographical locations underscores the framework's generalizability to different wind regimes and atmospheric conditions.

Alongside predictive accuracy, practical wind-speed forecasting also depends on computational cost and memory footprint. Table 5 shows that the superior accuracy of VORTEX is achieved with a larger model than most baselines, with 22.40 M parameters, 1.15 G forward-pass FLOPs, 10.80 ms inference time, and 89.59 MB of float32 parameter memory under the 24-hour input and 6-hour prediction setting. This increase is expected because the proposed framework combines regime-specialized recurrent computation with the DEM

Table 4
Comparative performance across datasets, averaged over 6-hour horizon.

Dataset	Metric	CNN BiGRU	ConvLSTM	CoReSTL	GBRN	PI LSTM	PIN TCN TFT	PatchTST	SZ TCN LSTM	**VORTEX**	iTransformer
Dodge City	RMSE	1.95	1.71	1.64	1.75	1.77	1.72	1.89	1.81	1.57	2.04
	MAE	1.47	1.24	1.19	1.31	1.29	1.27	1.43	1.36	1.12	1.55
	R^2	0.60	0.69	0.71	0.67	0.67	0.68	0.62	0.65	0.74	0.55
Jiuquan	RMSE	2.25	1.70	1.62	1.81	1.61	1.67	1.99	1.62	1.48	2.24
	MAE	1.75	1.23	1.17	1.35	1.13	1.24	1.54	1.17	1.06	1.75
	R^2	0.57	0.75	0.78	0.72	0.78	0.76	0.67	0.78	0.81	0.58
Pincher Creek	RMSE	1.95	1.15	1.13	1.30	1.16	1.25	1.61	1.46	1.03	1.57
	MAE	1.47	0.84	0.83	0.98	0.85	0.93	1.24	1.10	0.75	1.22
	R^2	0.60	0.83	0.83	0.78	0.83	0.80	0.67	0.73	0.86	0.68
Sarmiento	RMSE	1.79	1.35	1.48	1.65	1.34	1.46	2.10	1.70	1.25	1.84
	MAE	1.35	0.95	1.09	1.22	0.94	1.07	1.59	1.27	0.89	1.40
	R^2	0.82	0.90	0.88	0.85	0.90	0.88	0.75	0.84	0.91	0.81

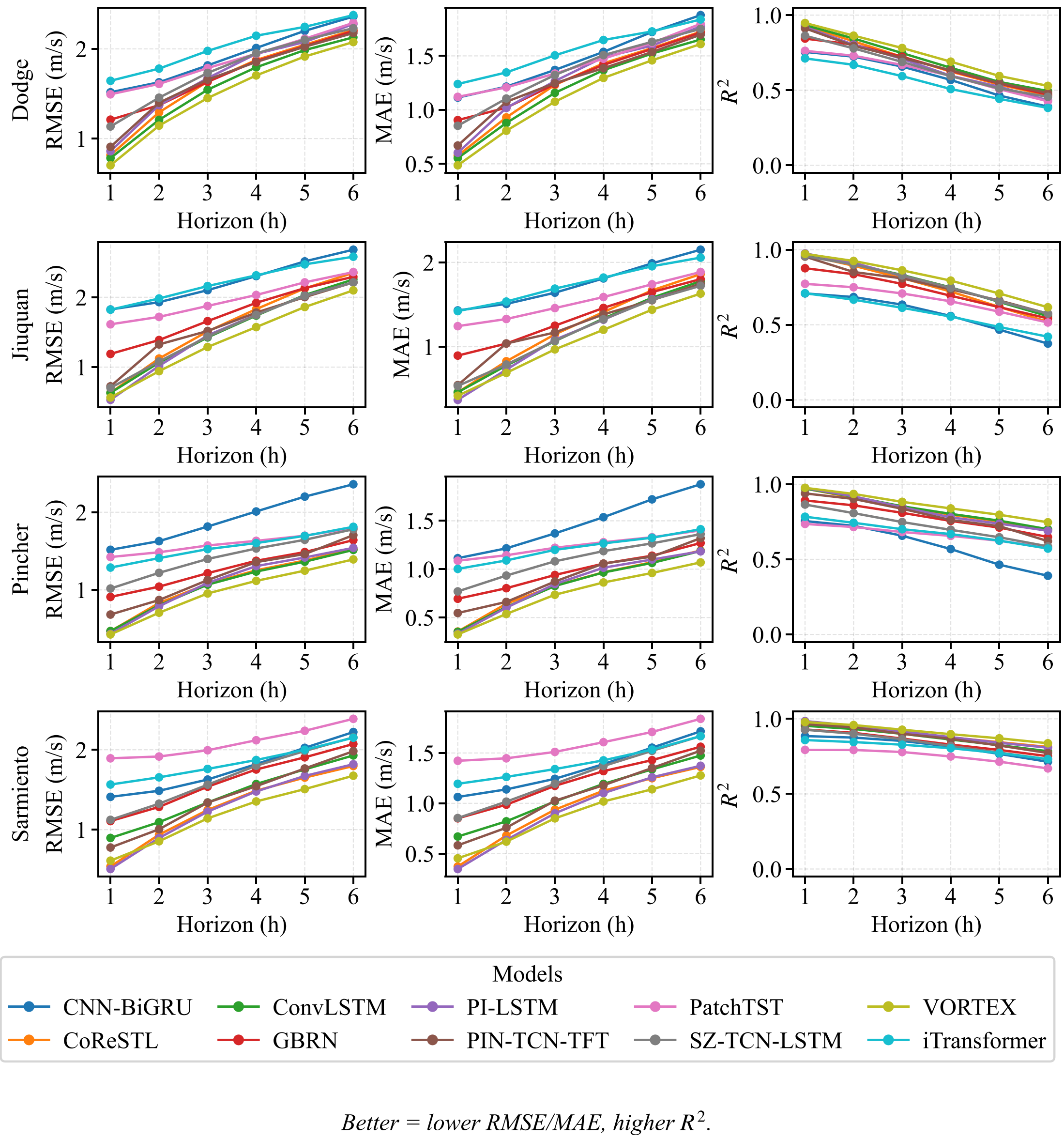


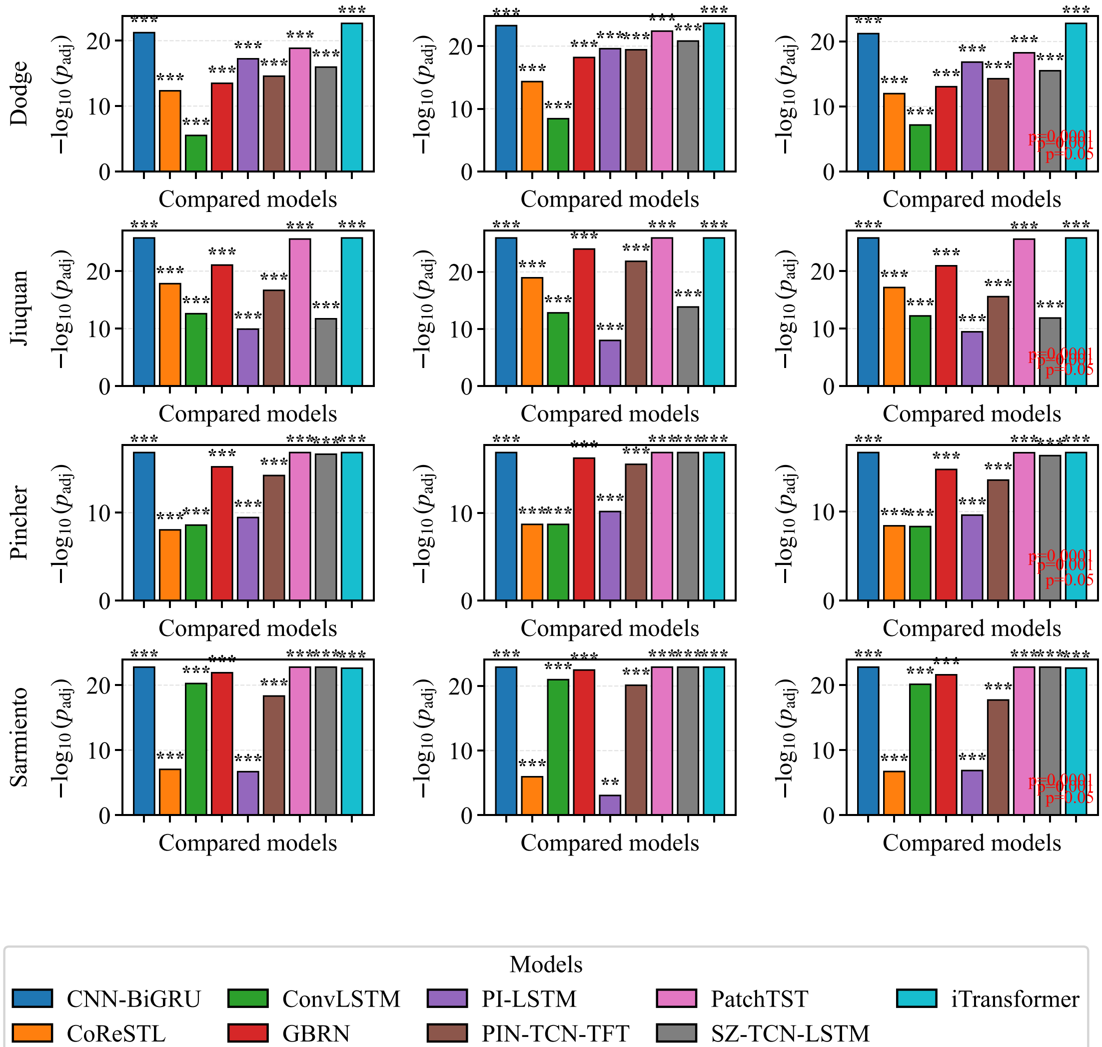
Fig. 6. Hour-by-hour forecast metrics across all datasets. Row 1: Dodge City, Row 2: Jiuquan, Row 3: Pincher Creek, Row 4: Sarmiento. Columns show RMSE, MAE and R^2 respectively.

feature-refinement block, both of which add representational capacity beyond a single shared backbone. However, the resulting complexity remains within a practical deployment range. In particular, VORTEX remains substantially less expensive than PatchTST in forward-pass computation and inference time, and it is also markedly lighter than ConvLSTM in computational complexity, while consistently delivering the best predictive performance in Table 4. These results indicate that the accuracy gains of VORTEX are obtained through an intentional but still manageable increase in model capacity, rather than through a computational burden that would preclude near-real-time operational use. In this sense, the proposed model favors predictive reliability under volatile wind regimes while remaining feasible for deployment on modern edge GPUs and embedded AI accelerators used in renewable-energy forecasting systems.

3.6. Volatility quartile evaluation

To comprehensively evaluate model performance across various atmospheric volatility regimes, this work conducts a quartile-wise analysis by partitioning the test set by wind speed volatility. The volatility for each test sample is the standard deviation of the actual wind speed over the 6-hour prediction horizon, computed from the reference ground truth. Samples are sorted by this volatility in m/s and divided into four equal-sized groups, denoted as quartile 1 (Q1), quartile 2 (Q2), quartile 3 (Q3), and quartile 4 (Q4), where Q1 contains the 25% least volatile cases and Q4 contains the 25% most volatile cases. For each quartile, the evaluation computes RMSE and MAE in m/s and R^2 as a unitless coefficient, which isolates behavior across stability conditions.

Fig. 8 compares VORTEX with the strongest baseline across volatility quartiles Q1 to Q4, where Q1 denotes the most stable regime and Q4 the most volatile. VORTEX leads in all quartiles for all datasets except Dodge in Q1, where ConvLSTM marginally outperforms it and the gain becomes slightly negative. Absolute performance for both VORTEX and the strongest baseline often degrades as volatility increases. This is consistent with the view that WSP accuracy declines under higher volatility. However, the relative improvement of VORTEX over the strongest baseline generally grows with volatility. Both models are affected by volatility, but VORTEX appears less sensitive, so its advantage widens



*Bars show $-\log_{10}(p_{\mathrm{adj}})$. Symbols: *** < 1e-4, ** < 1e-3, * < 5e-2, X ≥ 5e-2.*

Fig. 7. Statistical significance analysis using the Wilcoxon signed-rank test. Each row corresponds to a dataset (Dodge City, Jiuquan, Pincher Creek, Sarmiento) and each column corresponds to a metric (RMSE, MAE, R^2).

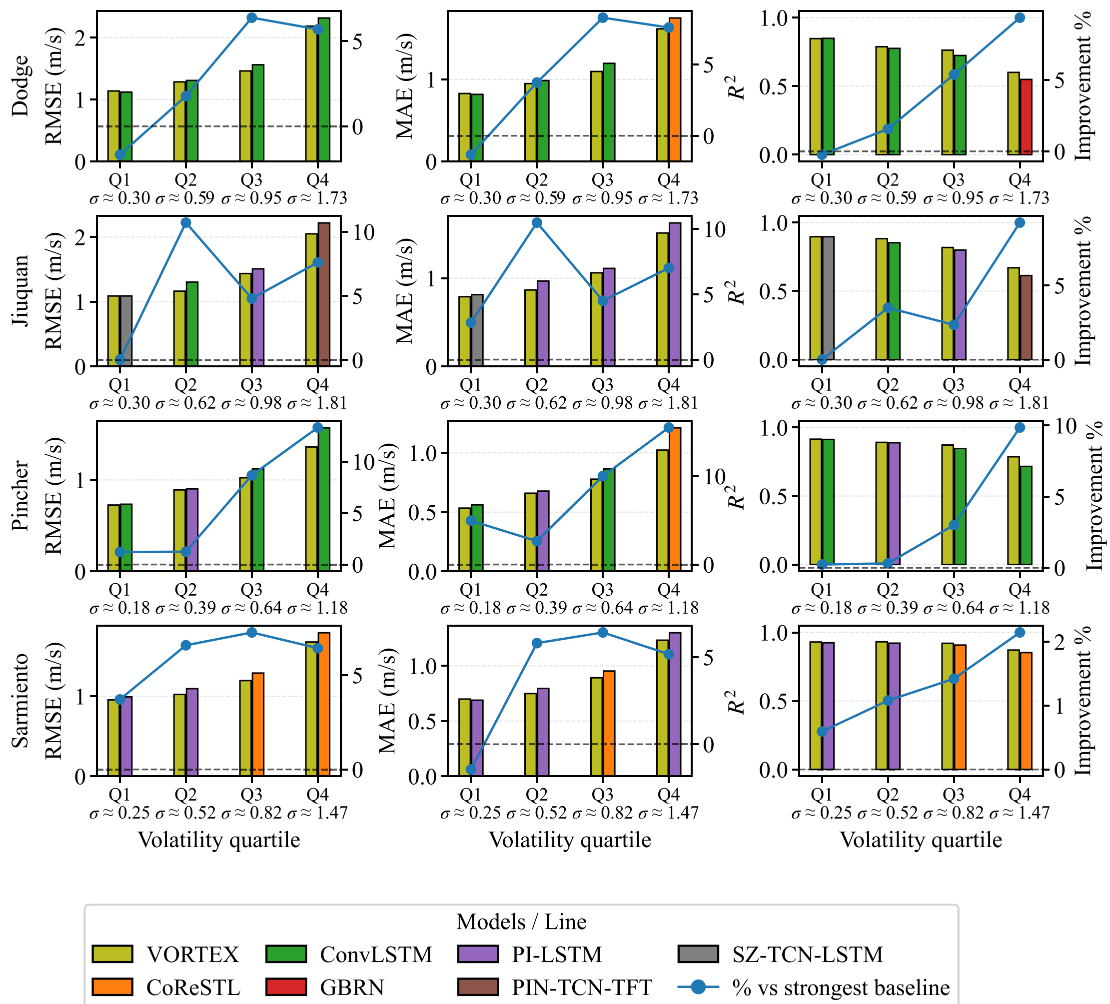
Table 5
Comparison of model size, forward-pass complexity, inference time, and parameter memory under the common 24-hour input and 6-hour prediction setting. Parameters are reported in millions (M), FLOPs in giga-floating-point operations (G), inference time in milliseconds (ms), and memory as float32 parameter memory.

Model	Parameters (M)	FLOPs (G)	Inference time (ms)	Memory (MB)
CNN-BiGRU	0.08	0.00	0.80	0.32
ConvLSTM	1.33	2.66	18.90	5.34
CoReSTL	0.44	0.36	5.90	1.77
GBRN	0.27	0.01	1.10	1.08
PI-LSTM	1.93	0.09	2.20	7.73
PIN-TCN-TFT	0.52	0.03	1.40	2.08
PatchTST	6.49	13.70	92.70	25.96
SZ-TCN-LSTM	4.78	0.22	4.10	19.13
VORTEX	**22.40**	**1.15**	**10.80**	**89.59**
iTransformer	2.42	0.12	2.50	9.69

in Q3 and Q4. Together, these results indicate that the performance advantage of VORTEX becomes more pronounced as volatility increases.

Table 6 reports detailed quartile-wise results across attention models, convolutional recurrent hybrids and physics-informed baselines. Errors increase from Q1 to Q4 in many sites and R^2 drops in parallel. Attention-centric models underperform across quartiles with large gaps in calm regimes. For example, in Pincher Q1 PatchTST reports 1.38 m/s while VORTEX reports 0.73 m/s and in Jiuquan Q4 iTransformer reaches 2.69 m/s while VORTEX reaches 2.05 m/s. Convolutional recurrent hybrids often provide the strongest baseline against VORTEX results in stable to moderate regimes. In Dodge Q1 ConvLSTM attains 1.12 m/s while VORTEX attains 1.14 m/s and in Pincher Q2 ConvLSTM records 0.93 m/s while VORTEX records 0.89 m/s. Physics-informed baselines become the strongest baselines against VORTEX models as volatility grows in several sites. In Sarmiento Q3 and Q4 PI-LSTM reports 1.33 m/s and 1.81 m/s and in Jiuquan Q4 PIN-TCN-TFT reports 2.22 m/s.

VORTEX matches or surpasses all baselines in every quartile except Dodge Q1 and its advantage widens as volatility increases. In Q4, the RMSE gain over the best baseline equals 5.70% in Dodge with 2.19 m/s vs. 2.32 m/s, 7.60% in Jiuquan with 2.05 m/s vs. 2.22 m/s, 13.30% in Pincher with 1.36 m/s vs. 1.57 m/s and 7.50% in Sarmiento with 1.68 m/s vs. 1.81 m/s. The pattern is consistent in R^2. In Dodge



Bars show $-\log_{10}(p_{\text{adj}})$. *Symbols: *** < 1e-4, ** < 1e-3, * < 5e-2, X ≥ 5e-2.*

Fig. 8. Quartile-wise comparison. Bars show absolute RMSE and MAE in m s^{-1} and R^2 (left axis). The line shows the % improvement of VORTEX over the strongest baseline (right axis). Quartile labels include the mean per-sample standard deviation $\bar{\sigma}$ in m s^{-1}.

Q4 VORTEX reaches 0.60 while the best baseline reaches 0.54 and in Pincher Q4 VORTEX reaches 0.79 while the best baseline reaches 0.72. These results indicate that attention baselines struggle with short-term WSP, hybrids excel when spatial coherence is strong, physics-informed models stabilize high-variance regimes and VORTEX remains superior across conditions because of its multi-gate routing, volatility-aware training and in-network refinement which learn regime-specific mappings rather than a single stationary solution.

3.7. Ablation study

This section reports an ablation study that isolates the contribution of each component in VORTEX. The analysis evaluates the effect of replacing the physics based volatility proxy with a statistical proxy, removing the DEM layer and replacing the volatility adaptive hybrid loss with standard MSE. All tables report Q4 RMSE so results align with the quartile analysis.

3.7.1. Physics-based vs. statistical volatility proxy

This ablation study examines the comparative efficacy of our physics-based volatility proxy against a conventional statistical alternative. We employ a rolling standard deviation as the statistical proxy, which quantifies volatility purely through historical fluctuation magnitude without physical grounding. The evaluation focuses on two representative sites: Sarmiento, representing a high-variance wind regime and Pincher Creek, representing a low-variance regime.

The physics-based proxy demonstrates consistent performance advantages across both regimes. It reduces Q4 RMSE by 1.47% in Sarmiento's high-variance environment and by 6.30% in Pincher Creek's low-variance setting compared to the statistical proxy. These improvements align with our earlier quartile analysis, where VORTEX exhibits the strongest performance gains during volatile conditions. The superior performance comes from fundamental differences in proxy construction. The statistical proxy operates as a post-hoc measurement of observed fluctuations, providing no causal insight into atmospheric dynamics. In contrast, the physics-based proxy quantifies the discrepancy between expected wind acceleration derived from pressure gradient forcing and actual observed changes. This approach captures the fundamental mechanical imbalance driving atmospheric instability, offering an interpretable measure that links volatility to specific physical processes rather than merely quantifying its magnitude. The physical grounding enables more accurate regime identification and consequently more effective model adaptation to changing atmospheric conditions.

Table 6
Quartile-wise performance across all datasets.

Dataset	Model	RMSE (m s^{-1})				MAE (m s^{-1})				R^2			
		Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Dodge	CNN-BiGRU	1.42	1.76	1.87	2.56	1.08	1.36	1.44	2.01	0.76	0.60	0.61	0.45
	CoReSTL	1.27	1.42	1.65	2.32	0.92	1.06	1.25	1.75	0.81	0.74	0.70	0.55
	ConvLSTM	1.12	1.31	1.56	2.32	0.82	0.99	1.20	1.75	0.85	0.78	0.72	0.54
	GBRN	1.28	1.50	1.74	2.32	0.95	1.15	1.34	1.80	0.81	0.71	0.66	0.55
	PI-LSTM	1.27	1.42	1.70	2.44	0.93	1.07	1.30	1.86	0.81	0.74	0.68	0.50
	PIN-TCN-TFT	1.28	1.39	1.66	2.35	0.92	1.06	1.27	1.82	0.81	0.75	0.70	0.54
	PatchTST	1.47	1.60	1.78	2.54	1.10	1.23	1.40	1.98	0.75	0.67	0.65	0.46
	SZ-TCN-LSTM	1.32	1.49	1.74	2.47	1.03	1.15	1.36	1.90	0.79	0.72	0.66	0.49
	VORTEX	1.14	1.29	1.46	2.19	0.83	0.95	1.10	1.62	0.85	0.79	0.76	0.60
	iTransformer	1.63	1.80	2.00	2.61	1.23	1.38	1.57	2.02	0.69	0.58	0.55	0.43
Jiuquan	CNN-BiGRU	1.81	2.03	2.15	2.86	1.44	1.63	1.72	2.23	0.72	0.64	0.59	0.36
	CoReSTL	1.28	1.43	1.61	2.30	0.96	1.05	1.19	1.71	0.86	0.82	0.77	0.59
	ConvLSTM	1.15	1.30	1.54	2.26	0.83	0.98	1.16	1.69	0.89	0.85	0.79	0.60
	GBRN	1.38	1.58	1.73	2.38	1.05	1.19	1.32	1.83	0.83	0.78	0.74	0.56
	PI-LSTM	1.15	1.34	1.51	2.22	0.83	0.97	1.11	1.63	0.89	0.85	0.80	0.61
	PIN-TCN-TFT	1.27	1.42	1.60	2.22	0.99	1.08	1.23	1.66	0.86	0.83	0.77	0.61
	PatchTST	1.70	1.78	1.95	2.43	1.31	1.39	1.54	1.92	0.75	0.72	0.66	0.54
	SZ-TCN-LSTM	1.09	1.32	1.55	2.27	0.81	1.00	1.17	1.68	0.90	0.85	0.79	0.59
	VORTEX	1.09	1.16	1.44	2.05	0.79	0.87	1.06	1.51	0.90	0.88	0.82	0.67
	iTransformer	1.95	1.99	2.24	2.69	1.54	1.56	1.75	2.13	0.67	0.66	0.56	0.43
Pincher	CNN-BiGRU	1.96	2.03	1.88	1.84	1.48	1.56	1.43	1.39	0.60	0.61	0.60	0.61
	CoReSTL	0.78	0.92	1.13	1.58	0.59	0.69	0.88	1.21	0.90	0.88	0.84	0.71
	ConvLSTM	0.73	0.93	1.12	1.57	0.56	0.70	0.86	1.21	0.91	0.88	0.85	0.72
	GBRN	0.93	1.15	1.33	1.68	0.70	0.89	1.03	1.31	0.86	0.82	0.78	0.67
	PI-LSTM	0.81	0.90	1.13	1.63	0.61	0.68	0.87	1.23	0.89	0.89	0.85	0.69
	PIN-TCN-TFT	0.80	0.99	1.22	1.77	0.62	0.77	0.96	1.38	0.90	0.87	0.82	0.64
	PatchTST	1.38	1.46	1.58	1.94	1.07	1.13	1.24	1.53	0.69	0.71	0.69	0.57
	SZ-TCN-LSTM	0.98	1.22	1.47	1.96	0.77	0.95	1.16	1.52	0.84	0.80	0.74	0.56
	VORTEX	0.73	0.89	1.02	1.36	0.53	0.66	0.78	1.02	0.92	0.89	0.87	0.79
	iTransformer	1.29	1.45	1.60	1.86	1.02	1.13	1.25	1.46	0.73	0.71	0.69	0.60
Sarmiento	CNN-BiGRU	1.39	1.52	1.76	2.34	1.05	1.19	1.36	1.80	0.85	0.85	0.83	0.75
	CoReSTL	1.03	1.14	1.29	1.79	0.73	0.83	0.95	1.30	0.92	0.92	0.91	0.85
	ConvLSTM	1.11	1.21	1.40	2.02	0.83	0.92	1.06	1.53	0.91	0.90	0.89	0.81
	GBRN	1.22	1.38	1.57	2.23	0.92	1.07	1.23	1.66	0.89	0.88	0.86	0.77
	PI-LSTM	0.99	1.10	1.33	1.81	0.69	0.80	0.96	1.30	0.92	0.92	0.90	0.85
	PIN-TCN-TFT	1.09	1.18	1.36	2.04	0.83	0.89	1.04	1.52	0.91	0.91	0.90	0.81
	PatchTST	1.74	1.86	2.02	2.65	1.32	1.41	1.58	2.04	0.77	0.78	0.77	0.68
	SZ-TCN-LSTM	1.22	1.43	1.67	2.28	0.94	1.12	1.27	1.75	0.89	0.87	0.85	0.76
	VORTEX	0.96	1.02	1.20	1.68	0.70	0.75	0.89	1.23	0.93	0.93	0.92	0.87
	iTransformer	1.53	1.67	1.79	2.29	1.18	1.30	1.39	1.74	0.82	0.82	0.82	0.76

3.7.2. *Physical relevance against a turbulence-related proxy*

To further examine the physical relevance of the proposed volatility proxy, this work compares V_{phys} against horizontal turbulent-kinetic-energy proxy defined as

$$\text{TKE}_{h,\text{proxy}} = \frac{1}{2}\left(\overline{u'^2} + \overline{v'^2}\right), \tag{35}$$

where $u' = u - \overline{u}$ and $v' = v - \overline{v}$ denote the rolling fluctuations of the horizontal wind components relative to their local window mean. This captures the intensity of horizontal velocity fluctuations and is commonly used as a surrogate for turbulence energy in situations where full three-dimensional measurements are unavailable.

Fig. 9 groups the samples into four equal-frequency bins ordered from low to high V_{phys} and reports the normalized mean $\text{TKE}_{h,\text{proxy}}$ in each bin for two representative sites that span contrasting variability regimes. In Pincher Creek, which exhibits relatively low variance, the normalized mean $\text{TKE}_{h,\text{proxy}}$ increases monotonically from Bin 1 to Bin 4 and reaches 1.94 times the Bin 1 level. This clear monotonic trend indicates that larger values of V_{phys} correspond to systematically stronger velocity fluctuations. In contrast, Sarmiento represents a higher-variance regime where atmospheric conditions are more irregular. In this case, the normalized mean $\text{TKE}_{h,\text{proxy}}$ shows a weaker but still positive increase, reaching 1.11 times the Bin 1 level. The reduced slope reflects the fact that, under highly variable conditions, multiple interacting processes contribute to wind fluctuations, which weakens a single-proxy relationship while preserving the overall trend. These results demonstrate that V_{phys} maintains a consistent large-scale association with turbulence-related fluctuation energy across distinct regimes.

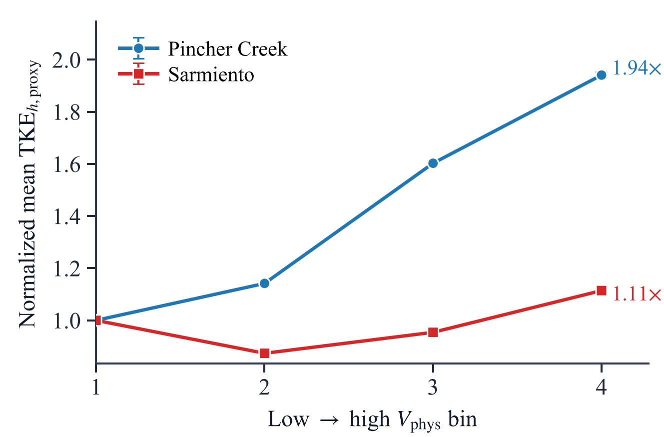


Fig. 9. Comparison between the proposed volatility proxy and a turbulence energy indicator. Samples are grouped into four equal-frequency bins ordered from low to high V_{phys}. The vertical axis shows the normalized mean $\text{TKE}_{h,\text{proxy}}$ within each bin.

Table 7
Ablation study: Physics-based versus statistical volatility proxy.

Dataset	Model	RMSE ($m\,s^{-1}$)	MAE ($m\,s^{-1}$)	R^2	Q4 RMSE ($m\,s^{-1}$)
Sarmiento	VORTEX (Statistical Proxy)	1.26	0.89	0.91	1.70
	VORTEX (Physics Proxy)	**1.22**	**0.86**	**0.92**	**1.68**
Pincher Creek	VORTEX (Statistical Proxy)	1.06	0.77	0.85	1.45
	VORTEX (Physics Proxy)	**1.05**	**0.76**	**0.86**	**1.36**

Table 8
Ablation study component contribution on Jiuquan.

Model	RMSE ($m\,s^{-1}$)	MAE ($m\,s^{-1}$)	R^2	Q4 RMSE ($m\,s^{-1}$)
VORTEX V1	1.67	1.20	0.76	2.36
VORTEX V2	1.52	1.09	0.80	2.16
VORTEX V3 (Full)	**1.48**	**1.06**	**0.81**	**2.05**

The proxy therefore does not merely reflect statistical variability but captures physically meaningful departures from pressure-gradient-driven flow. Together with the ablation results in Table 7 , this comparison supports the interpretation of V_{phys} as a physically grounded, coarse-resolution forcing-mismatch indicator rather than a purely statistical volatility descriptor.

3.7.3. Component contribution analysis

This section performs an ablation study that systematically isolates the contribution of each VORTEX component. Version 1 (V1) uses the physics proxy input and the MG-CRM backbone, whereas the second version (V2) adds the DEM layer to Version 1. Finally, Version 3 (V3) adds the proposed VAH-Loss to evaluate the end-to-end performance of the proposed model. To evaluate the incremental performance of these models, this work uses the Jiuquan dataset for comparison since its wind speed variance lies approximately in the middle of all the datasets.

The incremental performance gains in Table 8 demonstrate the complementary roles of each architectural component. Adding the DEM layer to the base architecture (V1 to V2) reduces RMSE by 8.93% and Q4 RMSE by 8.52%. This improvement stems from the DEM layer's capacity to refine feature representations in the final layers by adaptively suppressing uninformative features while enhancing discriminative patterns. The evolutionary mixture mechanism learns to weight different feature transformations optimally, effectively performing in-network feature selection without external preprocessing.

Further enhancement occurs with the integration of the VAH-Loss (V2 to V3), which reduces RMSE by an additional 2.48% and Q4 RMSE by 5.22%. The volatility-adaptive loss function dynamically regulates gradient updates based on regime membership, enabling specialized training of the MG-CRM weight sets for different volatility conditions. This regime-aware optimization ensures that each parameter subset receives appropriate gradient scaling proportional to its relevance to

Table 9
Short-term power results across datasets. The best value in each column per dataset is highlighted in green.

Dataset	Model	RMSE (MW)	MAE (MW)	nRMSE (%)	RMSE Q1	RMSE Q2	RMSE Q3	RMSE Q4
Dodge	VORTEX	0.29	0.18	14.72	0.23	0.24	0.28	0.40
	CNN-BiGRU	0.38	0.24	18.80	0.29	0.34	0.38	0.48
	CoReSTL	0.32	0.20	16.15	0.24	0.27	0.32	0.43
	ConvLSTM	0.31	0.19	15.49	0.21	0.25	0.30	0.43
	GBRN	0.33	0.21	16.53	0.24	0.28	0.34	0.43
	PI-LSTM	0.34	0.21	16.77	0.25	0.28	0.33	0.45
	PIN-TCN-TFT	0.33	0.21	16.61	0.25	0.28	0.33	0.44
	PatchTST	0.36	0.24	18.15	0.30	0.31	0.34	0.47
	SZ-TCN-LSTM	0.34	0.22	17.16	0.27	0.29	0.34	0.45
	iTransformer	0.38	0.25	18.99	0.31	0.34	0.37	0.47
Jiuquan	VORTEX	0.26	0.15	13.07	0.20	0.21	0.26	0.34
	CNN-BiGRU	0.41	0.26	20.28	0.31	0.37	0.40	0.52
	CoReSTL	0.29	0.17	14.45	0.22	0.24	0.29	0.38
	ConvLSTM	0.28	0.16	13.81	0.19	0.22	0.27	0.38
	GBRN	0.31	0.19	15.71	0.23	0.29	0.31	0.40
	PI-LSTM	0.28	0.16	13.75	0.20	0.23	0.27	0.37
	PIN-TCN-TFT	0.30	0.18	14.93	0.23	0.27	0.31	0.37
	PatchTST	0.37	0.23	18.48	0.32	0.35	0.38	0.42
	SZ-TCN-LSTM	0.28	0.16	14.15	0.20	0.24	0.29	0.37
	iTransformer	0.40	0.26	19.97	0.34	0.35	0.41	0.48
Pincher	VORTEX	0.17	0.09	8.54	0.12	0.17	0.17	0.22
	CNN-BiGRU	0.37	0.24	18.50	0.38	0.37	0.37	0.36
	CoReSTL	0.18	0.10	8.91	0.13	0.15	0.17	0.24
	ConvLSTM	0.18	0.10	9.00	0.12	0.16	0.18	0.24
	GBRN	0.22	0.13	10.83	0.16	0.20	0.22	0.27
	PI-LSTM	0.18	0.10	9.03	0.13	0.15	0.17	0.25
	PIN-TCN-TFT	0.19	0.12	9.68	0.13	0.16	0.19	0.27
	PatchTST	0.26	0.16	13.13	0.23	0.26	0.26	0.30
	SZ-TCN-LSTM	0.23	0.14	11.57	0.16	0.20	0.22	0.32
	iTransformer	0.25	0.15	12.28	0.23	0.24	0.25	0.27
Sarmiento	VORTEX	0.25	0.14	12.71	0.19	0.22	0.24	0.34
	CNN-BiGRU	0.35	0.21	17.41	0.27	0.30	0.34	0.45
	CoReSTL	0.27	0.15	13.37	0.19	0.23	0.25	0.36
	ConvLSTM	0.29	0.17	14.52	0.20	0.25	0.27	0.40
	GBRN	0.32	0.19	16.16	0.23	0.26	0.31	0.44
	PI-LSTM	0.26	0.14	13.11	0.18	0.22	0.25	0.36
	PIN-TCN-TFT	0.29	0.17	14.47	0.20	0.24	0.27	0.41
	PatchTST	0.38	0.25	19.17	0.32	0.35	0.38	0.47
	SZ-TCN-LSTM	0.31	0.19	15.74	0.22	0.27	0.30	0.43
	iTransformer	0.36	0.22	17.80	0.30	0.33	0.34	0.44

the current atmospheric state, leading to more effective specialization across volatility regimes.

The complete model (V3) achieves a 13.29% reduction in Q4 RMSE compared to the base configuration (V1), demonstrating that the components operate synergistically. The physics-based volatility proxy enables accurate regime identification, the MG-CRM provides specialized computational pathways, the DEM layer optimizes feature representation and the VAH-Loss ensures regime-proportional training emphasis. This integrated approach yields substantial performance gains, particularly during high-volatility conditions where conventional models typically struggle.

3.8. Power-space results and comparison

This work evaluates power forecasting performance by converting wind speed predictions to power output using a standard 2.00MW (MW) turbine power curve, with all metrics calculated over the 6-hour forecasting horizon. VORTEX demonstrates consistent superiority across all geographical locations, achieving the lowest RMSE, MAE, and normalized Root Mean Squared Error (nRMSE) values for every dataset as shown in Table 9. The framework reduces RMSE by 4.90% in Dodge City (0.31 MW to 0.29 MW), 5.00% in Jiuquan (0.28 MW to 0.26 MW), 4.20% in Pincher Creek (0.18 MW to 0.17 MW) and 3.10% in Sarmiento (0.26 MW to 0.25 MW), with corresponding MAE improvements of 5.80%, 6.10%, 8.00% and 2.90% respectively.

The multi-gated architecture enables specialized processing of volatility patterns, which is particularly evident in quartile analysis. While baseline models show occasional advantages in low-volatility conditions (Q1-Q2), VORTEX dominates in high-volatility regimes (Q3-Q4) where forecasting accuracy is most critical for grid operations. This performance advantage comes from the VAH-Loss function that dynamically adjusts error penalties across volatility regimes through its weighted combination of MAE, RMSE and MSE, prioritizing large-error suppression during extreme wind events. The integrated design enables superior handling of time-varying volatility patterns across diverse climatic conditions.

4. Conclusion

The VORTEX framework demonstrated that integrating physics-based volatility quantification with adaptive deep learning architectures substantially improved extreme volatile wind speed prediction. By grounding regime identification in atmospheric mechanics through pressure-gradient force discrepancy, the model established a causally interpretable foundation for volatility assessment that transcended purely statistical measures. The membership-gated architecture then used this physical insight to specialize computational pathways for different atmospheric conditions, enabling the model to maintain accuracy across calm and turbulent regimes without compromising either. Across the four geographically diverse datasets, VORTEX consistently achieved the best overall predictive performance over the 6-hour forecasting horizon against all compared baselines. The results showed that these gains were especially clear under highly volatile conditions, where regime-aware routing, DEM-based feature refinement, and VAH-Loss helped reduce large forecast errors more effectively than static architectures. In particular, the model reduced RMSE by 7.64% and MAE by 6.75% on Jiuquan relative to the strongest baseline, while the quartile-based analysis further showed that its relative advantage generally increased from calm to highly volatile regimes. This approach addressed the fundamental limitation of static architectures that either over-smoothed extreme events or amplified noise in stable conditions, providing a more nuanced and physically consistent forecasting capability.

The practical implications of this work extended to multiple aspects of renewable energy integration and grid management. The cubic relationship between wind speed and power generation meant that even small improvements in extreme volatile event prediction yielded substantial benefits for power system stability and reserve scheduling. By prioritizing high-volatility regimes through specialized parameter sets and adaptive loss weighting, VORTEX directly targeted the conditions that posed the greatest risk to grid operations. The framework's consistent performance across geographically diverse locations suggested robust generalizability to different climatic conditions and topographic settings. Furthermore, the integrated design eliminated dependency on external pre-processing stages, making the approach suitable for both operational forecasting systems and edge computing applications in distributed wind farms. At the same time, this study has several limitations. First, the experiments rely on hourly NASA POWER reanalysis data rather than turbine-level or plant-level SCADA measurements, which limits the resolution of local turbulence and operational control effects. Second, the physics-based volatility proxy is designed as a coarse large-scale forcing-mismatch indicator, so it does not explicitly resolve the full atmospheric momentum budget or sub-hour boundary-layer processes. Third, the power-space evaluation uses a single generic 2.00 MW turbine curve, which simplifies the diversity of real turbine characteristics and site-specific power conversion behavior. Future work will therefore extend the framework to higher-resolution atmospheric and SCADA datasets, evaluate site-specific turbine power curves, and investigate transfer learning and domain adaptation across locations, forecasting horizons, and other renewable-energy applications.

CRediT authorship contribution statement

Laeeq Aslam: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Runmin Zou:** Supervision, Project administration. **Yaohui Huang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Formal analysis, Conceptualization. **Gang Li:** Supervision, Conceptualization. **Fatima Yaqoob:** Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **Sara Mouafik:** Writing – review & editing, Writing – original draft, Visualization, Software. **Saad Yousaf:** Writing – review & editing, Writing – original draft, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data are available on request or from the NASA POWER Data Access Viewer at https://power.larc.nasa.gov/data-access-viewer/.

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